

A 4D Non-Associative Loop Algebra for Extraneous Solutions

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Abstract

This paper presents the formal algebraic construction of a 4-dimensional, power-associative, globally non-associative algebra over $\mathbb{R}$ spanned by the basis $\{1, i, t, it\}$. This system introduces a specialized radical unit defined by $\sqrt{t} = -1$, (t is currently a placeholder for this unit), designed to capture and evaluate multi-valued extraneous branches that standard associative fields structurally discard. We demonstrate that the multiplicative structure forms a closed Loop (non-associative magma), providing an analytic path-dependent framework viable as a cryptographic primitive for post-quantum key exchange networks.

1 Introduction and Split-Complex Number System Limitations

In standard complex and split-complex fields governed by global associativity, attempting to evaluate the negative root state of a radical unit forces a logical collapse. If an element j satisfies $j^2 = 1$, enforcing the radical condition $\sqrt{j} = -1$ immediately causes:

$$j = (\sqrt{j})^2 = (-1)^2 = 1 \quad (1)$$

This violates the foundational split-complex constraint $j \neq \pm 1$. Standard calculus resolves this by discarding extraneous radical roots. This framework proposes an alternative architecture by abandoning global associativity to preserve path history.

2 Foundational Axioms and Loop Structure

We define a 4-dimensional vector space spanned by the linearly independent basis $\{1, i, t, it\}$ subject to the following three axioms:

1. $t^2 = 1$

2. $t \neq \pm 1$
3. $t^{1/2} = -1$

The multiplicative operations are defined completely by the following non-associative table:

$\cdot$	1	i	t	it
1	1	i	t	it
i	i	-1	it	t
t	t	$-it$	1	$-i$
it	it	$-t$	i	-1

Evaluating the expression $i \cdot i \cdot t$ demonstrates the broken symmetry:

$$(i \cdot i) \cdot t = (-1) \cdot t = -t \quad (2)$$

$$i \cdot (i \cdot t) = i \cdot it = t \quad (3)$$

Since $-t \neq t$, global associativity is destroyed, protecting the definition of $\sqrt{t} = -1$.

For this reason, it is not a value of the imaginary unit i multiplied by t but a unit.

3 Analytic Consistency and Taylor Expansion

Applying the definition of continuous exponents $t^n = e^{n \ln(t)}$, the system yields the precise constraint $\ln(t) = 2\pi i$. Evaluating this via an infinite Taylor series confirms self-consistency:

$$t^n = \sum_{k=0}^{\infty} \frac{(2\pi i n)^k}{k!} = \cos(2\pi n) + i \sin(2\pi n) \quad (4)$$

For $n = 1/2$, $t^{1/2} = \cos(\pi) + i \sin(\pi) = -1$. For $n = 2$, $t^2 = \cos(4\pi) + i \sin(4\pi) = 1$. This demonstrates how the exponential logarithmic identity work to prove the first and third axioms. However this method is flawed for the hyper complex unit t as this would mean $t^1 = 1$, which violates the second axiom

4 Analytic Consistency and Non-Linear Phase Modulation

To establish a self-consistent continuous mapping across the discrete sheets of this non-associative geometry without inducing a structural collapse into

the base field, the continuous power function t^n cannot scale with a linear phase (e.g., $2\pi n$). Under global associativity, a linear mapping forces the identity $t^1 = e^{2\pi i} = 1$, which immediately violates our foundational axiom $t \neq \pm 1$ and collapses the 4-dimensional vector space into a 2-dimensional complex plane.

To bypass this continuous field collapse, we introduce a non-linear topological phase-modulation function $\theta(n)$. We define the continuous exponential representation of the structural basis unit t as a parameterized mapping into the $\{1, t\}$ subalgebra:

$$t^n = \cos(\theta(n)) + t \sin(\theta(n)) \quad (5)$$

To satisfy our three fundamental boundary constraints while maintaining a smooth, differentiable path across the Riemann sheets, $\theta(n)$ is uniquely determined as the lowest-degree Lagrange interpolation polynomial matching our axioms:

1. **Axiom 3 Boundary** ($n = 1/2$): $t^{1/2} = -1 \implies \theta(1/2) = \pi$
2. **Identity Boundary** ($n = 1$): $t^1 = t \implies \theta(1) = \frac{\pi}{2}$
3. **Axiom 1 Boundary** ($n = 2$): $t^2 = 1 \implies \theta(2) = 0$

Setting up the corresponding system of linear equations for the quadratic form $\frac{\theta(n)}{\pi} = An^2 + Bn + C$, we solve for the unique coefficients:

$$\frac{1}{4}A + \frac{1}{2}B + C = 1 \quad (6)$$

$$A + B + C = \frac{1}{2} \quad (7)$$

$$4A + 2B + C = 0 \quad (8)$$

Solving this system yields $A = \frac{1}{3}$, $B = -\frac{3}{2}$, and $C = \frac{5}{3}$. Thus, the topological phase-modulation function is explicitly defined as:

$$\theta(n) = \pi \left(\frac{1}{3}n^2 - \frac{3}{2}n + \frac{5}{3} \right) \quad (9)$$

Evaluating the system across its continuous limits confirms strict analytical self-consistency:

- For $n = 1/2$, $\theta(1/2) = \pi \implies t^{1/2} = \cos(\pi) + t \sin(\pi) = -1$.
- For $n = 1$, $\theta(1) = \frac{\pi}{2} \implies t^1 = \cos(\frac{\pi}{2}) + t \sin(\frac{\pi}{2}) = t$.
- For $n = 2$, $\theta(2) = 0 \implies t^2 = \cos(0) + t \sin(0) = 1$.

Because global associativity is destroyed, the exponent n does not function as a standard linear multiplier but rather tracks a non-linear path trajectory through the branched covering space. The quadratic variation of $\theta(n)$ acts as the exact algebraic correction needed to absorb the geometric distortion of the branch cut, ensuring that t preserves its independent coordinate identity ($t^1 = t$) throughout continuous evaluation.

4.1 Derivation of the Periodic Wave-Modulation Phase

However, to satisfy the constraint of local power-associativity across all integer iterations ($t^1 = t$, $t^2 = 1$, $t^3 = t$, $t^4 = 1$), the phase function $\theta(n)$ must be periodic and invariant under integer updates. This is because inputting $n = 4$ into the phase modulator doesn't result in 1, even though $t^2 \times t^2 = 1 \times 1 = 1$. This standard polynomial expansion diverges to infinity, introducing systemic geometric violations (such as $t^4 = -1$). To enforce strict cyclic power-associativity while preserving the fractional state $t^{1/2} = -1$, $\theta(n)$ is constructed via a two-component trigonometric interpolation.

4.1.1 Step 1: Constructing the Integer Gating Engine

We require a foundational wave function that maps any even integer to a phase of 0 (ensuring $\cos(0) = 1$) and any odd integer to a phase of $\frac{\pi}{2}$ (ensuring $\sin(\frac{\pi}{2}) = 1$, yielding t). The baseline function $\sin(\frac{\pi n}{2})$ alternates signs at higher odd intervals ($n = 3 \implies -1$). To enforce strict positivity, we square the function, yielding the integer gating engine:

$$\theta_{\text{integer}}(n) = \frac{\pi}{2} \sin^2\left(\frac{\pi n}{2}\right) \quad (10)$$

Evaluating Eq. (10) across all integers demonstrates complete boundary stability:

$$n \in \{2, 4, 8, \dots\} \implies \theta_{\text{integer}}(n) = 0 \implies t^n = 1 \quad (11)$$

$$n \in \{1, 3, 5, \dots\} \implies \theta_{\text{integer}}(n) = \frac{\pi}{2} \implies t^n = t \quad (12)$$

4.1.2 Step 2: Formulating the Fractional Correction Term

Evaluating the integer engine at the fractional boundary $n = 1/2$ yields:

$$\theta_{\text{integer}}\left(\frac{1}{2}\right) = \frac{\pi}{2} \sin^2\left(\frac{\pi}{4}\right) = \frac{\pi}{2} \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{\pi}{4} \quad (13)$$

However, Axiom 3 structurally demands $\theta(1/2) = \pi$ to satisfy the extraneous root boundary condition. The gating engine is deficient by a magnitude of precisely:

$$\Delta\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4} \quad (14)$$

To supply this correction without corrupting the integer boundary states, we introduce a localized modulation term using $\sin(\pi n)$. This function possesses the ideal topological property of vanishing identically on the set of all integers $\mathbb{Z}$ while capturing a maximum amplitude at $n = 1/2$:

$$\sin(\pi n) = 0, \quad \forall n \in \mathbb{Z} \quad (15)$$

$$\sin\left(\pi \cdot \frac{1}{2}\right) = 1 \quad (16)$$

4.1.3 Step 3: Synthesis and Final Closed-Form Representation

Superimposing the fractional correction term multiplied by our target amplitude discrepancy $\Delta\theta$ onto the primary integer engine, we isolate the complete analytic phase function:

$$\theta(n) = \frac{\pi}{2} \sin^2\left(\frac{\pi n}{2}\right) + \frac{3\pi}{4} \sin(\pi n) \quad (17)$$

Factoring out the global scalar π provides the final, copy-pastable closed-form representation implemented within Section 3:

$$\theta(n) = \pi \left[\frac{1}{2} \sin^2\left(\frac{\pi n}{2}\right) + \frac{3}{4} \sin(\pi n) \right] \quad (18)$$

By deploying Eq. (18), the algebra functions smoothly across continuous boundaries. The second term drops to absolute zero for any whole number exponent, anchoring the local power-associativity rules of the magma. Conversely, for fractional transitions, it introduces the exact boundary deformation required to natively absorb the tracking history of the multi-valued Riemann surface sheets.

5 Analytic Consistency and 4D Harmonic Trajectories

To construct a continuous representation for the exponential power map t^n that respects both the broken global associativity of our algebra and the distinct spatial properties of our 4D basis $\mathcal{B} = \{1, i, t, it\}$, the parameter n cannot scale within a flat 2D linear phase plane. If a traditional exponential mapping is enforced under global associativity, the identity $t^1 = e^{2\pi i} = 1$ inevitably springs a dimensional collapse, forcing $t = 1$, which invalidates our foundational axiom $t \neq \pm 1$.

Furthermore, as the underlying architecture transitions through fractional root states, it must natively exhibit the geometric constraints of multi-valued Riemann coverings. Because our third axiom states $t^{1/2} = -1$, a secondary extraction of the radical state naturally yields $t^{1/4} = \sqrt{-1} = i$.

This structural condition demands that the continuous trajectory of t^n actively leak out of the $\{1, t\}$ plane and traverse the standard imaginary axis i , completely justifying the 4-dimensional hypercomplex vector representation $z = a + bi + ct + dit$.

We resolve this by defining the continuous exponential power map t^n as a parameterized trajectory mapping smoothly across four distinct orthogonal harmonic wave channels:

$$t^n = f_1(n) \cdot 1 + f_2(n) \cdot i + f_3(n) \cdot t + f_4(n) \cdot it \quad (19)$$

5.0.1 Step 1: Establishing the Integer Gating Engine

To guarantee strict local power-associativity across all integer bounds ($t^1 = t, t^2 = 1, t^3 = t, t^4 = 1$), the baseline functions must alternate perfectly between the scalar identity 1 and the sheet generator t at whole numbers. Utilizing the complementary trigonometric identity $\cos^2(\alpha) + \sin^2(\alpha) = 1$, we establish our baseline integer gating engine:

$$f_{1,\text{base}}(n) = \cos^2\left(\frac{\pi n}{2}\right) \quad (20)$$

$$f_{3,\text{base}}(n) = \sin^2\left(\frac{\pi n}{2}\right) \quad (21)$$

For any even integer k , Eq. (29) yields 1 and Eq. (30) yields 0. For any odd integer m , Eq. (29) yields 0 and Eq. (30) yields 1. This successfully stabilizes the whole-number iterations.

5.0.2 Step 2: Calibrating the Half-Step Boundary

Axiom 3 dictates that at the half-step interval $n = 1/2$, the state must evaluate precisely to the real scalar -1 . Evaluating our baseline engine at $n = 1/2$ reveals an algebraic deficit:

$$f_{1,\text{base}}\left(\frac{1}{2}\right) = \cos^2\left(\frac{\pi}{4}\right) = \frac{1}{2} \quad \left[\text{Target: } -1 \implies \text{Deficit: } -\frac{3}{2} \right] \quad (22)$$

$$f_{3,\text{base}}\left(\frac{1}{2}\right) = \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2} \quad \left[\text{Target: } 0 \implies \text{Deficit: } -\frac{1}{2} \right] \quad (23)$$

To correct this boundary without corrupting our established integer coordinates, we introduce the harmonic filter $\sin(\pi n)$. This function possesses the ideal property of vanishing identically for all integers $n \in \mathbb{Z}$, while capturing a maximum unit amplitude at $n = 1/2$. Multiplying this filter by our deficits yields the stage-2 coordinates:

$$f_{1,\text{stage } 2}(n) = \cos^2\left(\frac{\pi n}{2}\right) - \frac{3}{2} \sin(\pi n) \quad (24)$$

$$f_{3,\text{stage } 2}(n) = \sin^2\left(\frac{\pi n}{2}\right) - \frac{1}{2} \sin(\pi n) \quad (25)$$

5.0.3 Step 3: Integrating the Quarter-Step Complex Leakage

At the quarter-step interval $n = 1/4$, the trajectory must land exactly on the imaginary unit axis i , requiring $f_2(1/4) = 1$, $f_1(1/4) = 0$, and $f_3(1/4) = 0$. For the imaginary coordinate $f_2(n)$, we introduce $\sin(2\pi n)$, which peaks perfectly at $\sin(\pi/2) = 1$ when $n = 1/4$, but collapses to absolute zero for all integers and half-steps, isolating the channel.

Next, evaluating our stage-2 real and sheet equations at $n = 1/4$ utilizing half-angle reductions reveals a shared spatial offset:

$$f_{1,\text{stage } 2}\left(\frac{1}{4}\right) = \left(\frac{2 + \sqrt{2}}{4}\right) - \frac{3\sqrt{2}}{4} = \frac{1 - \sqrt{2}}{2} \quad (26)$$

$$f_{3,\text{stage } 2}\left(\frac{1}{4}\right) = \left(\frac{2 - \sqrt{2}}{4}\right) - \frac{\sqrt{2}}{4} = \frac{1 - \sqrt{2}}{2} \quad (27)$$

To drive both coordinates to zero at $n = 1/4$, we require a higher-order harmonic patch with an amplitude of exactly $\frac{\sqrt{2}-1}{2}$. The wave function $\sin(2\pi n)$ provides the ideal topological profile, as it acts as a maximum correction at $n = 1/4$ but completely vanishes at all integer bounds $n \in \mathbb{Z}$ and the half-step boundary $n = 1/2$. Superimposing this term completes the full 4D system.

5.1 Closed-Form Coordinate Definitions

The explicit spatial distribution functions governing the trajectory across the 4D basis are defined as follows:

$$f_1(n) = \cos^2\left(\frac{\pi n}{2}\right) - \frac{3}{2}\sin(\pi n) + \left(\frac{\sqrt{2}-1}{2}\right)\sin(2\pi n) \quad (28)$$

$$f_2(n) = \sin(2\pi n) \quad (29)$$

$$f_3(n) = \sin^2\left(\frac{\pi n}{2}\right) - \frac{1}{2}\sin(\pi n) + \left(\frac{\sqrt{2}-1}{2}\right)\sin(2\pi n) \quad (30)$$

$$f_4(n) = 0 \quad (31)$$

5.2 Structural Topology Preservation (The Why)

This specialized multi-channel wave framework functions as an algebraic firewall. By separating the trajectory into independent spatial coordinates driven by orthogonal wave periods, the framework accomplishes three vital structural goals:

1. **Prevention of Dimensional Flattening:** Because the coordinate filters explicitly separate the basis units, the analytic identity $t^1 = t$

coexists perfectly alongside the numeric projection boundaries. The non-associative unit t is fiercely protected from collapsing into the real scalar number 1.

2. **Strict Local Power-Associativity:** Because every single fractional correction wave component ($\sin(\pi n)$ and $\sin(2\pi n)$) drops to absolute zero for any whole number integer exponent, the system guarantees that integer iterations remain completely locked within a stable, repeating associative loop ($t^1 = t, t^2 = 1, t^4 = 1, t^8 = 1$).
3. **True Path History Tracking:** The system acts as a physical trajectory tracker across a branched Riemann surface. It demonstrates that continuous exponentiation within a non-associative algebra isn't a static scalar state, but a dynamic movement that weaves systematically through real numbers, shifts entirely into standard imaginary numbers at quarter-steps, and blooms into the structural sheet generator t at whole numbers.

5.3 The General Hypercomplex Master Equation for t^z

To complete the analytical profile of our system, we extend the continuous path trajectory to accommodate a general 4D hypercomplex exponent variable $z = a + bi + ct + dit$. Because global associativity is broken and the standard imaginary basis i anti-commutes with the structural sheet unit t under magma multiplication ($t \cdot i = -it$), traditional exponent splitting laws (e.g., $t^{A+B} = t^A \cdot t^B$) cannot be assumed. Instead, the general exponential power function t^z must be solved by treating z as a split-complex variable over the standard complex domain, expressed as $z = Z_1 + Z_2 t$, where $Z_1 = a + bi$ and $Z_2 = c + di$.

By applying the hypercomplex extension mapping dictated by the idempotent property of our structural unit ($t^2 = 1$), the general master formula for t^z resolves uniquely into a vector sum over our 4D basis $\mathcal{B} = \{1, i, t, it\}$:

$$t^z = \mathcal{A} \cdot 1 + \mathcal{B} \cdot i + \mathcal{C} \cdot t + \mathcal{D} \cdot it \quad (32)$$

To compute the real scalar coordinate coefficients $\{\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$, our continuous coordinate wave filters $f_k(n)$ are evaluated at the complex boundary conjugate projections $z_+ = (a + c) + (b + d)i$ and $z_- = (a - c) + (b - d)i$.

This yields the finalized closed-form master system:

$$\mathcal{A} = \frac{1}{2}\text{Re}[f_1(z_+) + f_1(z_-)] - \frac{1}{2}\text{Im}[f_2(z_+) - f_2(z_-)] \quad (33)$$

$$\mathcal{B} = \frac{1}{2}\text{Im}[f_1(z_+) + f_1(z_-)] + \frac{1}{2}\text{Re}[f_2(z_+) - f_2(z_-)] \quad (34)$$

$$\mathcal{C} = \frac{1}{2}\text{Re}[f_3(z_+) + f_3(z_-)] - \frac{1}{2}\text{Im}[f_4(z_+) - f_4(z_-)] \quad (35)$$

$$\mathcal{D} = \frac{1}{2}\text{Im}[f_3(z_+) + f_3(z_-)] + \frac{1}{2}\text{Re}[f_4(z_+) - f_4(z_-)] \quad (36)$$

Here, $\text{Re}[f_k(z_{\pm})]$ and $\text{Im}[f_k(z_{\pm})]$ represent the standard real and imaginary parts of our coordinate functions after mapping their complex arguments onto their native hyperbolic forms ($\cosh$, $\sinh$).

5.3.1 Boundary Reductions and Systemic Consistency

The structural robustness of the master system is verified by analyzing its boundary collapses against our core axioms:

- **The Identity Zero-State** ($z = 0$): For $a = b = c = d = 0$, the complex projections collapse to $z_+ = z_- = 0$. Imposing this state into Eq. (49) forces $\mathcal{A} = f_1(0) = 1$, while all remaining channels vanish identically to zero, yielding $t^0 = 1$. This preserves the strict identity requirements of an algebraic Loop.
- **Purely Real Integer Trajectories** ($z = a \in \mathbb{Z}$): This state enforces $z_+ = z_- = a$. The difference matrices in Eqs. (49–52) collapse to zero, recovering our real multi-channel path $t^a = f_1(a) + f_3(a)t$. For even integers, this reduces cleanly to $t^{2k} = 1$, and for odd integers to $t^{2k+1} = t$, satisfying local power-associativity.
- **Asymmetric Dimensional Warping** ($z = \pm it$): Setting $z = \pm it$ sets $z_+ = \pm di$ and $z_- = \mp di$. The additions and subtractions across the coordinate splits explicitly execute the cross-multiplication properties of our magma table, generating the precise asymmetric hyperbolic coordinates for our conjugate pairs t^{it} and t^{-it} .

This master formula successfully bounds the continuous topological trajectory to the discrete rules of our non-associative algebra, confirming complete algebraic and geometric self-consistency.

5.4 The Natural Logarithm $\ln(t)$ as a 4D Spatial Velocity Vector

Having established a robust master system for t^z , we can now rigorously define the natural logarithm of our structural basis unit, $\ln(t)$, without inducing a catastrophic dimensional collapse. In our initial formulation, defining

$\ln(t) = 2\pi i$ via traditional linear field mappings inadvertently forced the identity $t = e^{2\pi i} = 1$, which flattened our 4-dimensional vector space back into the standard complex plane.

To preserve the independent topological identity of t ($t \neq 1$), the logarithm must be formalized within our non-associative architecture not as a flat scalar or complex value, but as an instantaneous velocity vector field acting on our 4D hypercomplex manifold. By utilizing the analytic definition of the exponential map, the logarithm represents the exact derivative of our continuous multi-channel trajectory evaluated at the identity origin state ($n = 0$):

$$\ln(t) = \left. \frac{d}{dn}(t^n) \right|_{n=0} = f'_1(0) \cdot 1 + f'_2(0) \cdot i + f'_3(0) \cdot t + f'_4(0) \cdot it \quad (37)$$

We compute the exact coordinate velocities by differentiating our established orthogonal harmonic wave filters with respect to the continuous path parameter n :

$$f'_1(n) = \frac{d}{dn} \left[\cos^2\left(\frac{\pi n}{2}\right) - \frac{3}{2} \sin(\pi n) + \left(\frac{\sqrt{2}-1}{2}\right) \sin(2\pi n) \right] \quad (38)$$

$$f'_2(n) = \frac{d}{dn} [\sin(2\pi n)] \quad (39)$$

$$f'_3(n) = \frac{d}{dn} \left[\sin^2\left(\frac{\pi n}{2}\right) - \frac{1}{2} \sin(\pi n) + \left(\frac{\sqrt{2}-1}{2}\right) \sin(2\pi n) \right] \quad (40)$$

Evaluating Eqs. (55–57) at the origin boundary $n = 0$ yields the precise spatial velocity weights across our 4D basis:

$$f'_1(0) = 0 - \frac{3\pi}{2} \cos(0) + 2\pi \left(\frac{\sqrt{2}-1}{2}\right) \cos(0) = \pi \left(\sqrt{2} - \frac{5}{2}\right) \quad (41)$$

$$f'_2(0) = 2\pi \cos(0) = 2\pi \quad (42)$$

$$f'_3(0) = 0 - \frac{\pi}{2} \cos(0) + 2\pi \left(\frac{\sqrt{2}-1}{2}\right) \cos(0) = \pi \left(\sqrt{2} - \frac{3}{2}\right) \quad (43)$$

$$f'_4(0) = 0 \quad (44)$$

Assembling these components into our hypercomplex representation yields the complete closed-form vector resolution for the logarithm of our structural unit:

$$\ln(t) = \pi \left(\sqrt{2} - \frac{5}{2}\right) \cdot 1 + 2\pi \cdot i + \pi \left(\sqrt{2} - \frac{3}{2}\right) \cdot t \quad (45)$$

Evaluating Eq. (62) numerically projects the logarithmic state onto the following constant 4D spatial coordinates:

$$\ln(t) \approx -3.4111 + 6.2831i - 0.2699t \quad (46)$$

5.5 Proof of Inverse Exponential Reversibility

To demonstrate complete systemic self-consistency, we verify that passing our derived 4D logarithmic velocity vector back into our exponential multi-channel mapping engine flawlessly reconstructs the native structural basis vector t :

$$e^{\ln(t)} = e^{-3.4111+6.2831i-0.2699t} = t \quad (47)$$

The geometric mechanism of this inverse mapping can be verified by analyzing how the orthogonal wave channels modulate under the specific vector components of $\ln(t)$:

1. **The Complex Rotation Channel:** The imaginary component $(f'_2(0)i = 2\pi i)$ represents a perfect 360° loop around the standard complex axis. Because $\sin(2\pi) = 0$ and $\cos(2\pi) = 1$, it strips away any residual complex leakage from the final state, ensuring the output collapses strictly onto the real-sheet interface.
2. **The Non-Associative Deflection Channels:** The real component (≈ -3.4111) and the sheet component $(\approx -0.2699t)$ function as precise geometric damping weights. Under our non-associative magma multiplication rules, these coordinates generate a controlled hyperbolic deformation that dampens the expanding waves of the space at exactly the right rate to force a total destructive interference across the $\{1, i, it\}$ coordinates.

When the wave components finish interfering constructively and destructively across the 4D manifold, the entire hypercomplex exponential expression resolves to:

$$e^{\ln(t)} = 0 \cdot 1 + 0 \cdot i + 1 \cdot t + 0 \cdot it = t \quad (48)$$

This mathematically proves that our continuous path trajectory is perfectly reversible. The non-associative 4D framework successfully supports continuous logarithms and exponentials while fiercely protecting the independent geometric identity of our structural unit ($t^1 = t \neq 1$).

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